SymPy Bug Card

Bug 29: Negative branch returned for a principal fractional power

Task. Solve over the real domain:

$$x^{2/3} = 4.$$

SymPy's answer.

$$\mathrm{solveset}(x^{2/3} = 4,\ x,\ \mathbb{R}) = \{-8, 8\}.$$

Correct answer.

$$\{8\}.$$

Explanation. The returned value -8 does not satisfy the original SymPy expression: under principal-power semantics,

$$(-8)^{2/3} = 4(-1)^{2/3} = -2 + 2\sqrt{3}\,i \neq 4.$$

The negative branch is valid for a real-root interpretation, but not for the principal `Pow` expression that was solved.

Likely root cause. The diagnosis traces the result to `_invert_real` in `sympy/solvers/solveset.py`, whose rational-power branch adds both $y^{3/2}$ and $-y^{3/2}$ for exponent $2/3$. That inverts real-root semantics rather than principal `Pow` semantics, making this a new instance of a known failure mode.

Artifact (GitHub). `bug-report/bug-029-negative-branch-returned-for-a-principal-fractional-power/`